The descent formula for preorder polytopes

MathDB 374958

Principia Math In Progress

September 18, 2026

Abstract

For every preorder on an arbitrarily totally ordered n-element set, we prove that the h^*-polynomial of its preorder polytope is the reverse-descent enumerator of words meeting every order ideal at least as often as its cardinality. This proves Conjecture 4.4 of Athanasiadis and Chapoton. We use bivariate Ehrhart reciprocity, their established counting formula, and an explicit multiset argument.

Task 1. Full solution

Let E be a finite set with $|E| = n \geq 1$, equipped with a preorder τ and, separately, an arbitrary total order. There is no compatibility assumption between these two orders. Set

$$\mathcal{Q}_\tau = \left\{ x \in \mathbb{R}^E_{\geq 0} : \sum_{e \in I} x_e \leq |I| \text{ for every order ideal } I \right\}.$$

Write τ^* for the reversed preorder, and $P_{\tau^*} = \mathcal{Q}_{\tau^*} \cap \mathbb{Z}^E$. Let W_τ consist of the words $w \in E^n$ such that the letters from each order ideal I occur at least $|I|$ times in w. The number $\mathrm{des}(w)$ counts indices $1 \leq j < n$ with $w_j > w_{j+1}$ in the chosen total order.

Theorem 1. *For every such preorder and total order,*

$$\boxed{h^*(\mathcal{Q}_\tau, t) = \sum_{w \in W_\tau} t^{n-1-\mathrm{des}(w)}.}$$

A counting identity from reciprocity

We give the needed reciprocity argument in full. Let

$$\mathcal{Q}_\tau(r, s) = \left\{ x \in \mathbb{R}^E_{\geq 0} : \sum_{e \in I} x_e \leq r|I| + s \text{ for every nonempty order ideal } I \right\}.$$

By [1, Propositions 3.2–3.3 and 7.1], for nonnegative integers r, s this is a lattice polytope, also expressible as an integral Minkowski sum of simplices. Its lattice-point count is the polynomial $\mathbb{E}_\tau(r, s)$, of total degree n.

For positive integers r, s, the polytope is full dimensional (it contains $[0, r]^E$). Its interior integer points satisfy $x_e \geq 1$ and $\sum_{e \in I} x_e \leq r|I| + s - 1$ for every nonempty order ideal. Subtracting the

all-ones vector therefore bijects these points with $\mathcal{Q}_\tau(r-1,s-1)\cap\mathbb{Z}^E$. All inequality normals are nonzero; redundant inequalities have no effect on this strict-interior characterization.

For $P=\mathcal{Q}_\tau(r,s)$ one has $kP=\mathcal{Q}_\tau(kr,ks)$ for nonnegative integers k, so its Ehrhart polynomial equals $\mathbb{E}_\tau(kr,ks)$ as a polynomial in k. Ordinary Ehrhart reciprocity at $k=-1$ and the translation just described imply

$$\mathbb{E}_\tau(-u,-v)=(-1)^n\mathbb{E}_\tau(u-1,v-1). \tag{1}$$

Initially this holds for positive integer pairs (u,v); it is a polynomial identity because a polynomial vanishing on that entire grid is zero (first fix v and vary u, then vary v).

For $a=(a_e)_{e\in E}$ write $|a|=\sum_e a_e$ and $a_0=n-|a|$. The established formula [1, Lemma 4.1 and Equation (25)] is

$$\mathbb{E}_\tau(u,v)=\sum_{a\in P_{\tau^*}}\binom{v+a_0}{a_0}\prod_{e\in E}\binom{u+a_e-1}{a_e}. \tag{2}$$

Here $a_0\geq 0$, and binomial coefficients with fixed nonnegative lower argument are interpreted as polynomials in the upper argument. Putting $(u,v)=(-m-1,-1)$ in (2), every term with $a_0>0$ vanishes, since $\binom{a_0-1}{a_0}=0$. For the remaining terms $|a|=n$, use

$$\binom{a_e-m-2}{a_e}=(-1)^{a_e}\binom{m+1}{a_e}.$$

Together with (1), this gives, for all $m\geq 0$,

$$\mathrm{Ehr}(\mathcal{Q}_\tau,m)=\mathbb{E}_\tau(m,0)=\sum_{\substack{a\in P_{\tau^*}\\ |a|=n}}\prod_{e\in E}\binom{m+1}{a_e}. \tag{3}$$

This implication of bivariate reciprocity was already noted, conditionally, in [1, Remark 7.3]; equation (1) supplies the missing premise.

An elementary multiset identity

For a nonnegative integer vector a with $|a|=n$, let S_a be all words having a_e copies of each e. We prove

$$\sum_{m\geq 0}\left(\prod_e\binom{m+1}{a_e}\right)t^m=\frac{\displaystyle\sum_{w\in S_a}t^{n-1-\mathrm{des}(w)}}{(1-t)^{n+1}}. \tag{4}$$

Choose, for each letter e, a subset of a_e bins from the $m+1$ ordered bins $0,1,\ldots,m$. Place one copy of e in each chosen bin. The number of choices is the product on the left. Within each bin, read its distinct letters in strictly decreasing order, and concatenate the bins. This produces a word $w\in S_a$ and a weakly increasing sequence of bin indices $0\leq b_1\leq\cdots\leq b_n\leq m$.

At each index j for which $w_j\leq w_{j+1}$, the bin index must strictly increase. Conversely these conditions suffice: a constant bin segment is strictly decreasing and contains no repeated letter. Thus this construction is a bijection. The number of required strict increases is $c(w)=n-1-\mathrm{des}(w)$. Subtract from b_j the number of required increases before position j. When $m\geq c(w)$ this gives a weakly increasing sequence of length n in $\{0,\ldots,m-c(w)\}$, counted by $\binom{m-c(w)+n}{n}$. There are no such sequences for $m<c(w)$. Consequently the contribution of w to the generating function is

$$\sum_{m\geq c(w)}\binom{m-c(w)+n}{n}t^m=\frac{t^{c(w)}}{(1-t)^{n+1}},$$

which proves (4), including repeated letters and $m = 0$.

Proof of Theorem 1. A vector a of total sum n belongs to P_{τ^*} exactly when $\sum_{e\in J} a_e \le |J|$ for every order filter J of τ. Taking complements $I = E \setminus J$, these conditions are equivalent to $\sum_{e\in I} a_e \ge |I|$ for every order ideal I of τ. Therefore the sets S_a indexed in (3) partition W_τ. Sum (4) over those vectors and use (3). The definition

$$\sum_{m\ge 0} \mathrm{Ehr}(\mathcal{Q}_\tau, m)t^m = \frac{h^*(\mathcal{Q}_\tau, t)}{(1-t)^{n+1}}$$

now yields the desired identity. □

Verification and attribution

Athanasiadis and Chapoton supplied the counting formula and proposed the implication from reciprocity. The complete reciprocity proof is reproduced above from our companion solution to MathDB 374965. The elementary multiset argument makes the descent conversion explicit. The earlier result [2] treats a special family of arbors. The later preorder result in [3, Corollary 5.4] concerns the reflexive polytope $\mathcal{Q}_\tau + \Delta_E - \mathbf{1}$, whose word set includes an extra letter; it is a different statement.

All 26 pages of the current primary were read. The accompanying verifier compares direct lattice counts with word enumeration for all 389 labeled preorders of sizes 1–4, checks 50,069 words for the multiset identity, and tests all 120 total orderings in the source's five-element example. Bounded checks found no prior full target claim or identified journal final. Review is by the authoring agent only, without independent peer review or a proof assistant.

Tasks 2–5. Proposed follow-up directions

Task 2. Construct an explicit bijection directly from lattice points of $m\mathcal{Q}_\tau$ to the bin assignments in (3).

Task 3. Find a geometric realization of descent-set refinements of the formula, retaining the positions of descents rather than their number.

Task 4. Derive colored-word analogues for integral weights assigned to the principal-filter simplex summands.

Task 5. Characterize extremal h^*-coefficients among preorders of fixed size using the ideal constraints on the word set.

References

[1] C. A. Athanasiadis and F. Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916v1, 2026.

[2] C. A. Athanasiadis, Q. Xiao, and X. Yan, *Lattice point enumeration of some arbor polytopes*, arXiv:2603.11654v1, 2026.

[3] X. Yan, *Variations of colored multiset Eulerian polynomials and applications*, arXiv:2608.15682v2, 2026.